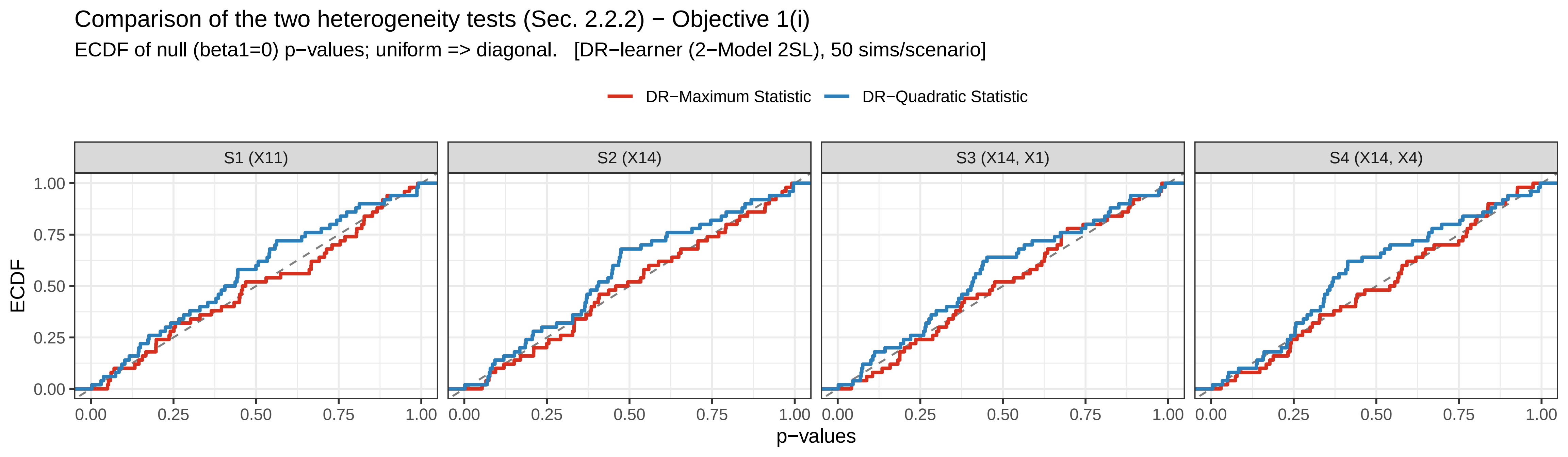
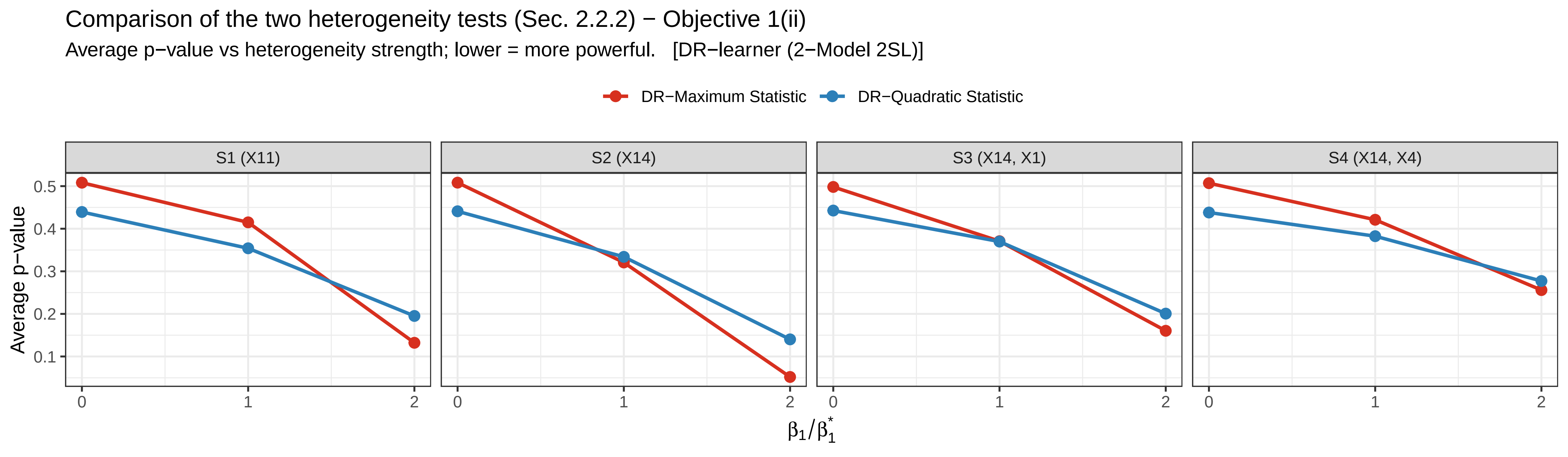
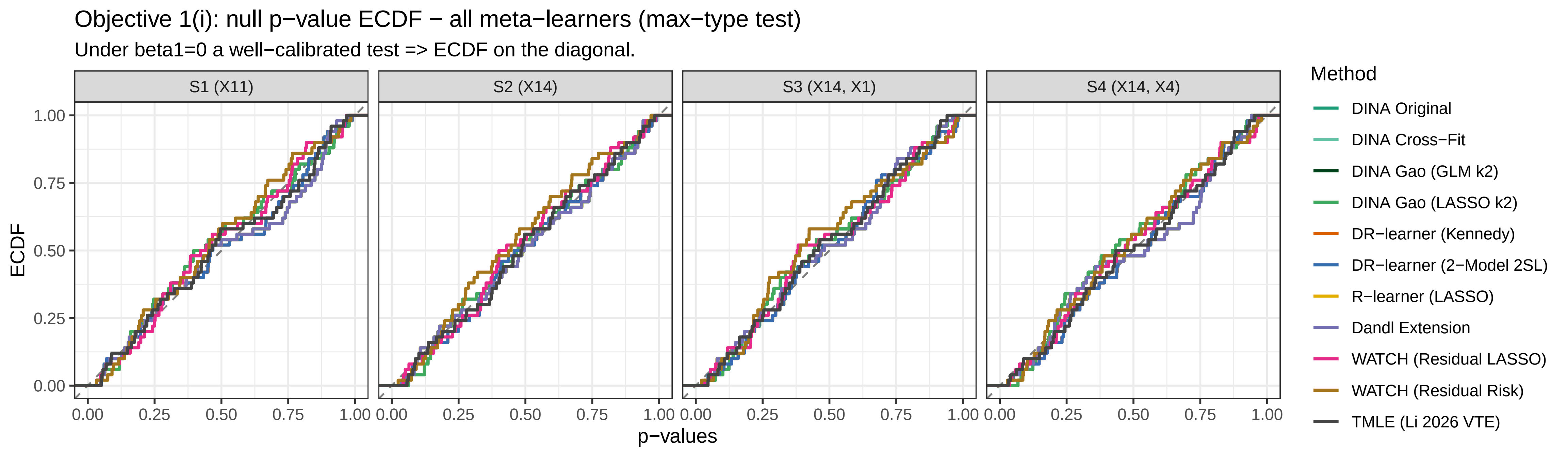
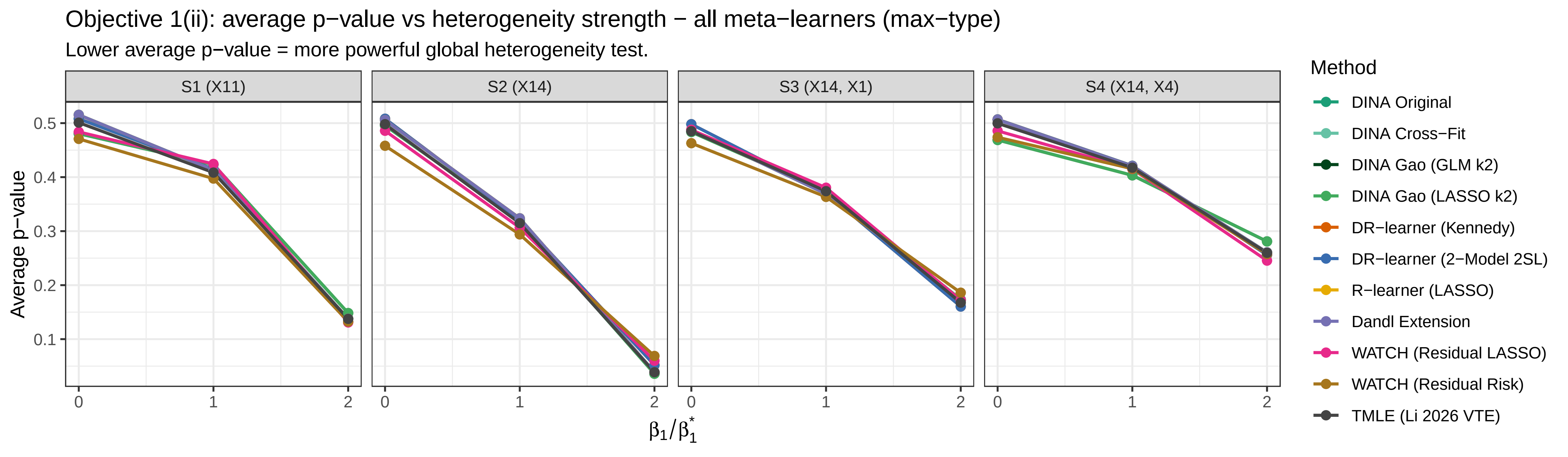

Comparison of the two heterogeneity tests (Sec. 2.2.2) − Objective 1(i)
ECDF of null (beta1=0) p−values; uniform => diagonal. [DR−learner (2−Model 2SL), 50 sims/scenario]
DR−Maximum Statistic
DR−Quadratic Statistic
S1 (X11)
S2 (X14)
S3 (X14, X1)
S4 (X14, X4)
ECDF
p−values
0.00
0.25
0.50
0.75
1.00



Comparison of the two heterogeneity tests (Sec. 2.2.2) − Objective 1(ii)
Average p−value vs heterogeneity strength; lower = more powerful. [DR−learner (2−Model 2SL)]
DR−Maximum Statistic
DR−Quadratic Statistic
S1 (X11)
S2 (X14)
S3 (X14, X1)
S4 (X14, X4)
Average p−value
0.1
0.2
0.3
0.4
0.5
0
1
2
β_1/β_1^*


Objective 1(i): null p−value ECDF − all meta−learners (max−type test)
Under beta1=0 a well−calibrated test => ECDF on the diagonal.
S1 (X11)
S2 (X14)
S3 (X14, X1)
S4 (X14, X4)
ECDF
p−values
0.00
0.25
0.50
0.75
1.00
Method
DINA Original
DINA Cross−Fit
DINA Gao (GLM k2)
DINA Gao (LASSO k2)
DR−learner (Kennedy)
DR−learner (2−Model 2SL)
R−learner (LASSO)
Dandl Extension
WATCH (Residual LASSO)
WATCH (Residual Risk)
TMLE (Li 2026 VTE)



Objective 1(ii): average p−value vs heterogeneity strength − all meta−learners (max−type)
Lower average p−value = more powerful global heterogeneity test.
S1 (X11)
S2 (X14)
S3 (X14, X1)
S4 (X14, X4)
Average p−value
0.5
0.4
0.3
0.2
0.1
0
1
2
β_1/β_1^*
Method
DINA Original
DINA Cross−Fit
DINA Gao (GLM k2)
DINA Gao (LASSO k2)
DR−learner (Kennedy)
DR−learner (2−Model 2SL)
R−learner (LASSO)
Dandl Extension
WATCH (Residual LASSO)
WATCH (Residual Risk)
TMLE (Li 2026 VTE)